

Gesture Recognition Based on MediaPipe for Computer Game Control

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ABSTRACT

The social function of computer games has gradually been recognized in modern society. With the rapid development of computing science, games based on human-computer interaction (HCI) and computer vision have attracted people's attention. Unlike traditional games based on keyboard or controller, these games focus on the user experience, where user interaction can create an immersive feeling that is beyond the reach of graphical user interfaces. This paper researches the application of gesture recognition technology in games via Google Mediapipe framework and Pygame. We evaluate the performance through hand gesture recognition and Flappy bird game control. The experiment shows the good result and user experience.

KEYWORDS

Computer vision; Mediapipe; Game control

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1 Introduction

In modern society, the social function of computer games has gradually been recognized. In addition to pure entertainment, computer games play an important role in releasing people's tensiety and life pressures. With the rapid development of computing science, games based on human-computer interaction (HCI) and computer vision have attracted people's attention. Unlike traditional games based on keyboard or controller, these games focus on the user experience, where user interaction can create an immersive feeling that is beyond the reach of graphical user interfaces. The Natal program launched by Microsoft on Xbox 360 has become a global sensation, providing a new gaming method that allows users to track motion, recognize faces, and even listen to the voice without the need for a controller.

Due to the important role played by gesture interaction in HCI, the application of computer vision technology in hand gesture recognition has become a hot research spot in many fields, such as medical field, virtual surgery field, animation field, language teaching field, intelligent robot field, etc. Although gesture recognition technology is not yet fully mature, it has been tried to be applied as an effective way of HCI in people's daily life. The application of gesture recognition technology in games mainly has two aspects. One is the somatosensory game, which breaks through the operation mode of simple handle key input, used on computers or television terminal devices. The other is touch screen games which use touch sensors installed in front of the display screen to detect the touch information of the user's fingers, converting the position and movement information of the fingers into contact coordinates to achieve game control.

In this paper, we research the application of gesture recognition technology in games. The direct

use of human gesture game control is more consistent with people's cognitive and behavioral habits. The way of gesture control games is simpler and easier for beginners to learn. Compared to the physical gaming mode, the gesture recognition in games does not require additional hardware devices, which reduces the cost of the game.

2 Literature Review

Computer vision based gesture recognition uses cameras to capture gesture images, and then uses computer vision technology to analyze the captured images, extract gesture image features, and achieve gesture recognition. Lee and Kunni^[1] used stereo image data to automatically analyze 3D gestures. They captured hand motion images using a camera and used contour extraction boundary features for recognition. 27 interactive hand parameters were successfully extracted and the reconstruction of 3D gestures were achieved. Since then, computer vision based gesture recognition has made great progress. By using the edge feature pixels of the hand area as the recognition feature, and using the idea of Hausdorff distance template matching, Liangguo Zhang, Jiangqin Wu, Wen Gao, et al.^[2] successfully realized Gesture recognition of 30 Chinese Sign Language finger letters based on gloveless monocular vision in the distance transformation space. Bousesmart et al.^[3] used simple gestures to manipulate several movable pie shaped control menus for immersive interaction in CAVE environments. Hand gestures are recognized via visual based video tracking technology. Semantic mapping is achieved between gestures and scene responses through pie shaped transparent controls. This approach enables a simple set of two handed gestures to support a variety of different interactive effects.

Hand recognition is the basis of gesture recognition. Due to the noises such as background clutter, background changes, and occlusion, hand recognition is full of challenges. At present, gesture segmentation methods are mainly divided into monocular vision based and stereo vision based methods. Hillebrand et al.^[4] used an infrared optical system to track fingers. A cloth glove was used to assist in the tracking of 3D gestures^[5]. Wang et al.^[6], on the other hand, adopted a compromise between wearable motion capture devices and bare hand tracking, using a colored glove (divided into 12 out of 10 colors) for hand pose estimation, achieving real-time human hand motion tracking.

Once the hand is successfully separated from the background, gesture recognition can be performed. Usually, the motion trajectory of the hand is extracted first, and then the gesture is recognized by analyzing the motion trajectory^[7-9].^[10] two kinds of Gesture recognition algorithms are summarized: rule-based method and machine learning method. In rule-based methods, different gesture rules are first derived, and then used to design different gestures. Ross Cutler et al.^[11] designed a gesture set using a rule-based approach. However, due to the different gesture habits of each individual, setting up a universal gesture set is not an easy task. In machine learning based methods, hidden Markov models (HMMs) are commonly used to train datasets^{[12][13]}. Wong et al.^[14] used an adaptive sparse Bayesian classifier recognized 10 basic gestures. However, methods based on hidden Markov models typically involve large-scale training datasets, which would limit their widespread application.

3 Gesture Recognition Based on Mediapipe

Google has implemented a gesture recognizer based on the Mediapipe framework, which utilizes machine learning to infer 21 3D hand keypoints within a single frame. As shown in the figure 1, the

framework provides high-fidelity tracking trajectories for hands and fingers. The framework is trained on approximately 30K real-world images, as well as several rendered synthetic hand models imposed over various backgrounds.

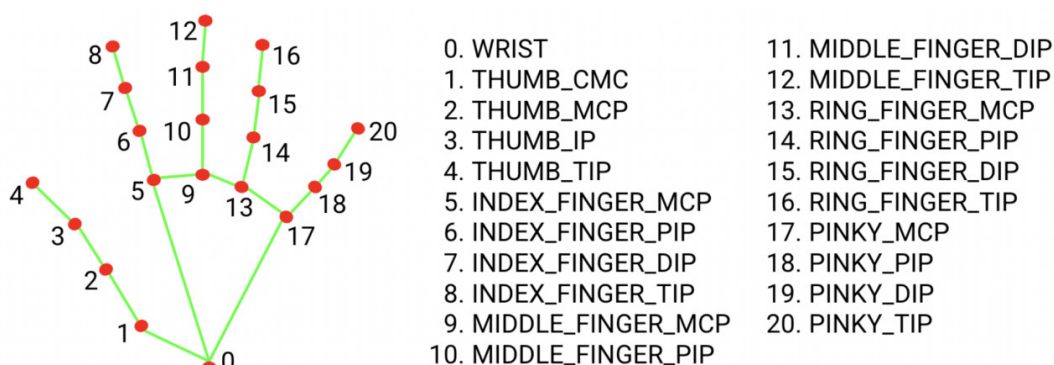


Figure 1 hand landmark from Mediapipe

The Mediapipe based gesture recognizer contains palm detection model and hand landmarks detection model. Palm detection model localizes the region of hands from the whole input image, and the hand landmarks detection model finds the landmarks on the cropped hand image defined by the palm detection model.

Since palm detection model is much more time consuming, in Video mode or Live stream mode, Gesture recognizer uses bounding box defined by the detected hand landmarks in the current frame to localize the region of hands in the next frame. This reduces the times Gesture Recognizer triggering palm detection model. Only when the hand landmarks model could no longer identify enough required number of hands presence, or the hand tracking fails, palm detection model is invoked to re-localize the hands. The architecture of the Gesture recognition system is shown as figure 2.

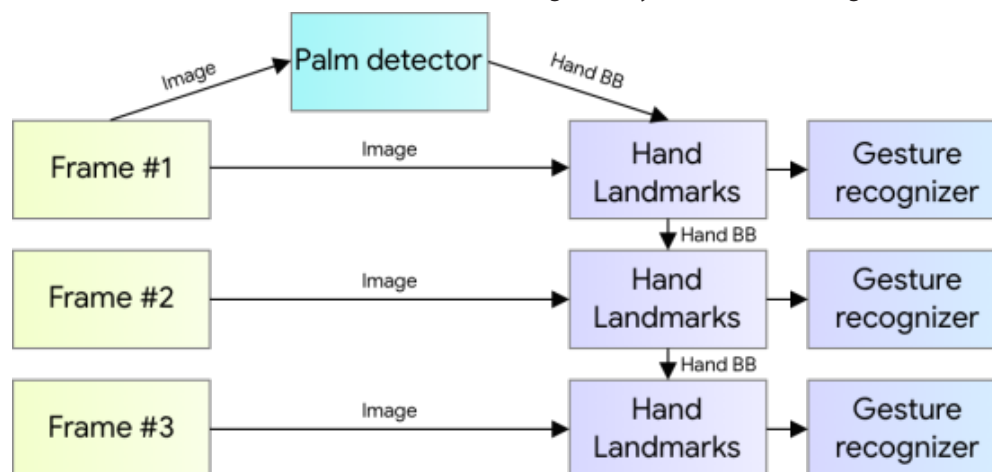


Figure 2 The architecture of the gesture recognition system

4 The Proposed Least Squares Fitting Algorithm for Linear Gesture

To further extract the semantics of gesture interaction, the discrimination of gesture movement direction is needed. Is a linear gesture a horizontal gesture, a vertical gesture, or a tilt gesture? Does it move from left to right or from right to left, or from bottom to top or from top to bottom? By

combining information such as the start point, end point, and inclination of linear gestures, the above questions can be easily answered.

To obtain the inclination of a linear gesture, this paper proposes the least squares method to perform linear fitting on the discrete sampling points of the gesture, and then characterizes the inclination of the linear gesture via the slope of the fitted line.

The functional relationship between gesture sampling points $P_i(x_i, y_i), i = 1, 2, \dots, n$, x , and y is given by the linear equation (1), where the two undetermined parameters b_0 and b_1 represent intercept and slope respectively.

$$y = b_0 + b_1 x \dots\dots\dots(1)$$

When using the proposed least squares algorithm for line fitting, the best estimate is made for parameter b_0 and b_1 , and the sum of the sampling values y_i is required to be the smallest.

$$\min \sum_{i=1}^N [y_i - (b_0 + b_1 x_i)]^2 \Big|_{b=\hat{b}} \dots\dots\dots(2)$$

According to (2), we have

$$\frac{\partial}{\partial b_0} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2 \Big|_{b=\hat{b}} = -2 \sum_{i=1}^n (y_i - \hat{b}_0 - \hat{b}_1 x_i) = 0 \dots\dots\dots(3)$$

$$\frac{\partial}{\partial b_1} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2 \Big|_{b=\hat{b}} = -2 \sum_{i=1}^n (y_i - \hat{b}_0 - \hat{b}_1 x_i) x_i = 0 \dots\dots\dots(4)$$

Then, we have as below.

$$\begin{cases} \hat{b}_0 n + \hat{b}_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \\ \hat{b}_0 \sum_{i=1}^n x_i + \hat{b}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i \end{cases} \dots\dots\dots(5)$$

To solve the equations (5), the best estimates of line parameters b_0 and b_1 can be obtained via $\hat{b}_0$ & $\hat{b}_1$.

$$\hat{b}_0 = \frac{\left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i \right) - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n x_i y_i \right)}{n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2} \dots\dots\dots(6)$$

$$\hat{b}_1 = \frac{n \left(\sum_{i=1}^n x_i y_i \right) - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2} \dots\dots\dots(7)$$

$\hat{b}_1$ is the best estimate of the inclination of linear gestures.

5 Computer Game Developed Via Pygame

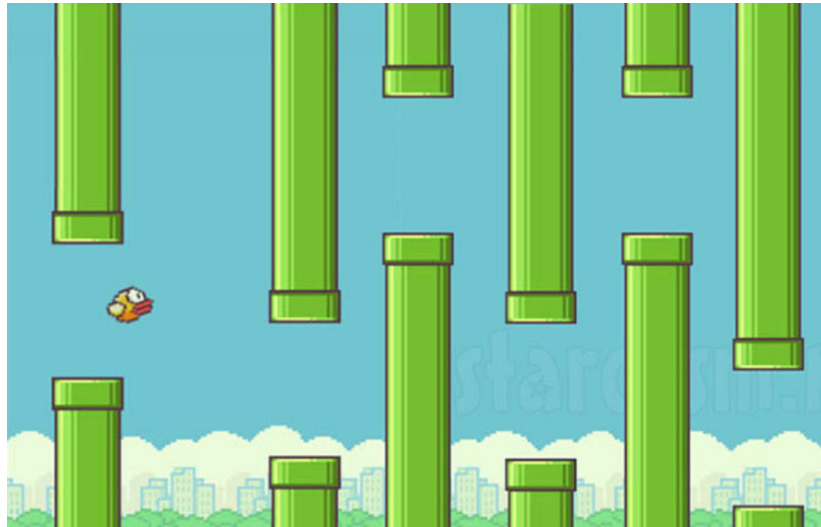
Pygame is a cross platform Python module for game development. Due to its SDL based design, Pygame has many multimedia related functions. For Python programs, using the Pygame module can create feature-rich games and a variety of multimedia programs, and because the Pygame module is partially developed in C language, it does not execute slowly. It also has the object-oriented, scripting, and high portability features of Python language.

When developing games using the Pygame module, each control is a relatively independent surface object. The function of a Surface object is to display images in Pygame, which can draw one image above another and set the transparency of the image. And the Rect class in Pygame can handle rectangular objects, such as text box controls, Button controls, and so on. The Rect class can define the parameters of components, making it easier for developers to use the attributes of images to complete the positioning layout in the container.

6 Flappy Bird Development

Flappy Bird is an arcade game where users control a likeable bird that has to fly through many obstacles all made up of pipes. Figure 3 shows the UI of Flappy bird.

The mechanics are very simple: users have to tap the screen so that the bird flaps its wings, trying to keep a steady rhythm in order to pass through the pipes scattered through its path. It won't be at all easy to do. This paper re-develop the classic game Flappy bird based on Pygame, and controls the game by the proposed gesture recognition algorithm through Mediapipe.



7 Experiment and Result

The algorithm proposed in this paper is implemented via Python on computer with Intel Core i5-2410M CPU and Nvidia GeForce GT550M GPU, and can also be executed on any other PC with Python installed.

Items	Parameters
CPU	Intel Core i5-2410M

GPU	Nvidia GeForce GT550M
Python	3.7.6
MediaPipe	0.9.0.1
PyGame	2.5.0

In this paper, gesture recognition is completed by a single hand and a monocular camera. The game can be controlled by changing the hand gesture. For example, expanding the palm indicates that the bird is flying upwards, while bending the palm indicates that the bird is flying downwards. Please refer to Figure 4 for specific gestures.



Figure 4 The example of expand & bend the palm

We invite 5 classmates to play the game. Each classmate would do at least 20 times of the gesture. To determine whether hand movements can be recognized and displayed normally, and to verify whether gestures and game controls correspond correctly, we use accuracy, precision, recall, F1-score as verification criteria.

Table1: Evaluation Criteria

Evaluation Criteria	Note
TP	True Positive: A true positive sample which is classified as positive
FP	False Positive: A true negative sample which is classified as positive
TN	True Negative: A true negative sample which is classified as negative
FN	False Negative: A true positive sample which is classified as negative
Accuracy	$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$
Precision	$Precision = \frac{TP}{TP + FP}$
Recall	$Recall = \frac{TP}{TP + FN}$
F_1 value	$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$

The test result is show as Figure 5.

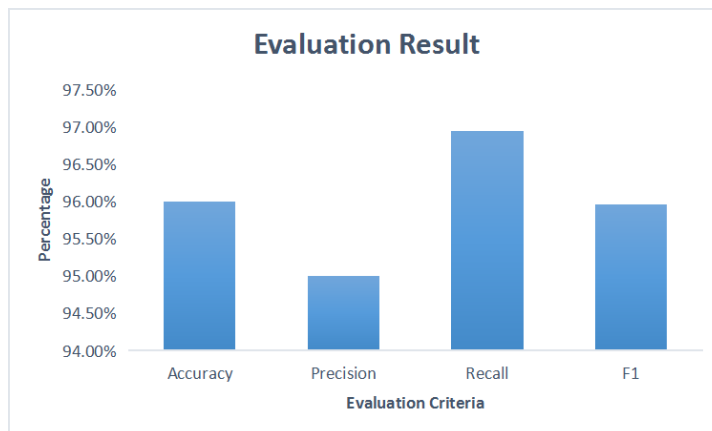


Figure 5 The result of our evaluation

From the result, we find the proposed algorithm have good performance, especially in accuracy and recall value. One reason is that Mediapipe is mature enough in single hand gesture recognition. The other reason is that the Flappy bird game control is relatively simple, only two gestures needed.

8 Conclusion

This paper mainly research hand gesture recognition based on Mediapipe, which is used for computer game control. The final function is to control the Flappy bird game through the recognized gesture, and two gestures are designed to complete the bird upward and downward control. The system's feasibility and robustness have been verified, which can accurately recognize gestures and control games, providing an idea for game operations through computer vision technology.

The limitation of this study is that there is no in-depth research on gesture recognition, and the number of gestures designed is small. In the future, we plan to research and develop more complex gesture control game operations.

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